NAME: ______________________ PERIOD: ________ DATE: ____________

Two Displays, One Data Set

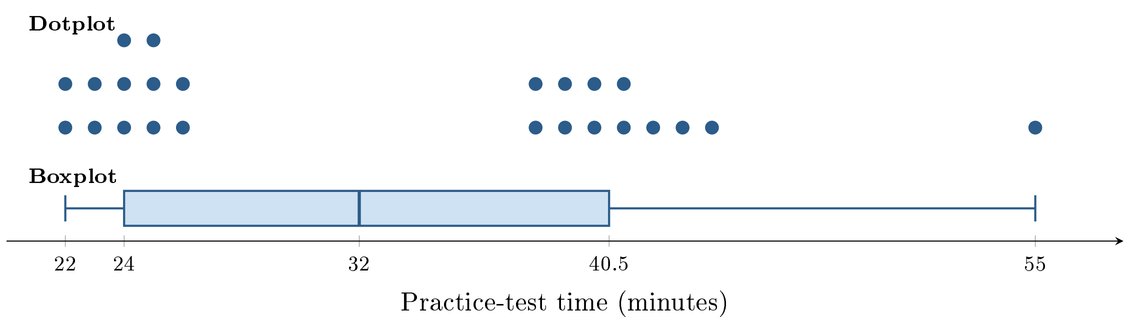
AP Statistics · Topic 1.8 (CED 1.8) · B: 2026-09-22 · E: 2026-09-28

[STAR] TODAY'S PROBLEM

The displays show practice-test times, in minutes, for 24 students. Two students interpret the boxplot.

Omar: "Most students took about 32 minutes—the middle of the box."

Nia: "A boxplot cannot tell us whether most students took about 32 minutes."



FIRST TAKE — before the video (5 min)

Who do you trust more, Omar or Nia? Point to one feature in either display.

[BOOK] DOTPLOTS AND BOXPLOTS (keep on the page)

A **dotplot** keeps every observation, so it can show clusters, gaps, peaks, and isolated values. A **boxplot** compresses the data to the five-number summary: minimum, Q1, median, Q3, maximum (with separately marked outliers). The box contains the middle 50% of observations. A boxplot shows center and spread efficiently, but it may hide modality, clusters, and gaps.

Finish after the video:

DOK 1 **(a)** Read the median and the first and third quartiles from the boxplot. What percent of the times lie in the box from Q1 to Q3?

DOK 2 **(b)** Describe the dotplot in context: include the two clusters, the gap, and the isolated high value. State two features that the boxplot preserves.

DOK 3 **(c)** Whose interpretation is better supported? Cite the dotplot feature that settles it. Explain what a boxplot can and cannot show here. A coach asks whether most students need more than 30 minutes: choose a display, and explain what the boxplot alone would lead the coach to believe.

Frame: ____________ is better supported because the dotplot shows ______________________________ around the median of 32 minutes.

Frame: The boxplot shows ____________________, but it hides _________________________, so the coach should use the ____________.

EXIT — turn this in

One thing the video changed about my first take: ____